

Subject Name: Specification, Estimation and Costing		
Course Contents & Hourly break-up		
Module Number	Topics	Number of hours
Module 1	Introduction to Specification	
	Definition importance and use of Specifications. Principles and Procedures of Specifications writing. Role of Specifications, Relation between Drawings and Specifications, Technical Section of Specifications, Different Types of Specifications, Specification Writing Techniques, Specification Language. Specification Clauses. General and Special Clauses. Contract Documents, Bidding requirement, Bonds and Warranties.	10
Module 2	Specification of different building materials	
	Specification of common building materials; viz. Bricks, Sand, Lime, Cement, Reinforcements. etc.	6
Module 3	Specification of different building items	
	Specifications of simple construction; viz: Earthwork in Excavation and Filling in Foundation Trenches, Masonry and RCC works. Different types of Flooring, Roof Treatments, Internal and External Finishing, Painting to Woodwork and Ironwork. Specifications of BIS and other Institutions. Specifications of new Building Materials.	8
Module 4	Introduction to Estimation and Costing	
	Purpose and types of estimates. Main items and their unit of measurement. Methods of Measurement and estimating quantities.	3
Module 5	Estimating quantities of Building	
	Estimation of quantity of load bearing and framed building structure. Estimation of quantity single storied residential building, Estimation of quantity Different R.C.C. structures. Assignments on small and medium size projects for application of these methods.	6
Module 6	Cost Estimation	
	Schedule of rates. Rate analysis: factors and procedure. Market survey methods. Wages of labour, labour and material requirement for different works, rate analysis of different Items of work. Cost calculation.	4